

1

03

medium



Scanned by CamScanner

২০ পৌষ সোমবার ১৪২১
৭৭ পৌষ শুক্ল সোমবার ২০৩৭
পূর্ণিমা ধাঃ ৯/৪১
Hizri - 13 Rabi-ul-awwal 1436

1G: $\rightarrow$ mobile comm - 50 miles
- wired

1G: $\rightarrow$ mobile comm - 50 miles
- wired

- availability of channel was less.

- 3 access techniques.

FDM

- less no. of users as less availability of channel.

- Then frequency re-use system came in 1G.

- Bcz of this no. of users ↑.

- FDM was used

- 2G: TDM (Signal processing technique)

- efficiency of Channel
- provide a speed of 10 kbps.

- 3G: Took 10 years 2G to 3G
- CDMA

- CDMA

২১ পৌষ মঙ্গলবার ১৪২১
৭ মাঘ কৃষ্ণ মঙ্গলবার ২০৪৭
প্রতিপদ ঘ: ১০/৫২
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January 06
Tuesday

- part. user part. code given.
- no signal interference

 (in TDM) (FDM)

↓
free slot for not interference
(Guard Band)

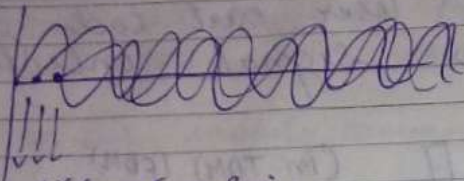
- Signal Changed to code

- Pseudo random code
given to every user.

Diagram showing a horizontal beam of length 12 ft, supported by a pin support on the left and a roller support on the right. A uniformly distributed load of 10 k/ft is applied downwards over the entire length of the beam.

100 kbps

- 4G: used OFDM
or orthogonal freq. DM
↓
no interferences



different origin.

(frequency bands doesn't interfere signal)

- Guard bands can use by other users in OFDM
- 100 mbps speed & also depends on geographical area
- But it's 1-2 Mbps
- 5G : formulated no. of signal processing

- 4G-10G
1. FBML (filter bank multi-carrier)
 2. SEFDM - (spectrally efficiency FDM)
 3. UFMC - (ultra filter multi-carrier)

JAN 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

✓ ⑪

- each technique hv their own advantage.

29/7/2017

Transmitting/receiving EM wave
c represent our voice/data through free space is wireless comm.

- antenna - receives EM wave
- Trans of data/voice - specific band - channel
- e.g GSM phone.

need 200KHz BW to transmitting for 1 user but for many users req. of BW is more and BW that huge is not available

- Only hv 25MHz available

JAN 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

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Id-milad-ul nabi

Hizri -17 Rabi-ul-awwal 1436

3. Portable comm

→ Mobile — every where
common any time
Slogan

Hizri-18-19 Rabi-ul-awwal 1436

Saturday

10

located in EM spectrum

Sunday 11

IEEE — 2.4, 5.4, 3.3
GHz

12 January (6)
 ২৭ পৌষ সোমবার ১৪২১
 ৬ মাঘ কৃষ্ণ সোমবার ২০৬৭
 সপ্তমী রাঃ ১০/২৭
 Janmochi / Janmochi of Swami Vivekananda Hizri - 20 Rabi-ul-awwal 1436

- Ultra WBand — police network
- body area network
- Band — 6.2–9.5 GHz

- Advantage of Wireless:
 comm — 3 stages.

1. Physical layer com
2. Microwave eng.
3. Computer Science

how we optionally use the freq & which freq used & how we transmit & design antenna.

- ↓ - as is a hill task:
- 1) - freedom from wires
 - 2) - we prefer it in

২৮ পৌষ মঙ্গলবার ১৪২১
 ৮ মাঘ কৃষ্ণ মঙ্গলবার ২০৬৭
 অষ্টমী রাঃ ১১/২৭
 Hizri - 21 Rabi-ul-awwal 1436

January 13
 Tuesday

emergency situatn e.g in adhoc networks (disasters situatn) i.e in situatn to install a cable.

- 3) We can achieve the last mile mission i.e how we will be able to give service to every household. (85)

- advantage are at the cost of challenges.
- cost per user ↓ as no. of users ↑

→ Challenges in achieving wireless comm.

- 1) HW problem
 - to design a mobile phone, it needs a battery. i.e to ↑ the battery life.

16 January Friday

1 माघ शुक्रवार १४३३
११ माघ कृष्ण शुक्रवार २०७१
अवधौ मा: १२/०१
Hizri - 24 Rabi-ul-awwal 1436

- as voice comm. is real time comm. so delay shd be zero.
- voice - circuitary switching
↓
(cannot tolerate delay) dedicated link established
- data - delay cannot be tolerable as packet switching.

(3) Bit error rate/ratio

- when we want quality of service
- B_r - Q.O.S.

JAN 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

latency - delay

२-३ माघ शनि-रवि १४३३
१२-१३ माघ कृष्ण शनि-रवि २०७१
अवधौ मा: १२/०१/१४३३
Hizri - 25-26 Rabi-ul-awwal 1436

January 17 Saturday

- how many bits we transmit from S to R
 - how many bits trans. wrongly?
- In voice,

lower value (achieve any case)

$$BER = 10^{-3}$$

only error in 1 bit

In data,

$$BER = 10^{-6}$$

Sunday 18

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 FEB 2015

4-8-2017

(8)

19

January
Monday

৪ মাঘ (শ্রাবণ) ১৪২১
১৭ মাঘ কৃষ্ণা সপ্তমী ১৪২১
উজ্জ্বলী রাি ১/১১
Hizri - 27 Rabi-ul-awwal 1436

Cellular Concept :-

Traditional telephone system a wire was used.

- maximum use of it could not be enough for the users as its limitation is wired, no long dist covered.

- Long dist — more noise

- RF Techn. came into existence (radio freq.)
- In it a single user was given a dedicated RF loop.
- Single RF loop requires B/w of 50 KHz.

- e.g 100000 subscribers create single RF loop

$$\text{So } 100000 \times 50 = 50 \text{ GHz}$$

So not possible.

৪ মাঘ (শ্রাবণ) ১৪২১

১৭ মাঘ কৃষ্ণা সপ্তমী ১৪২১

উজ্জ্বলী রাি ১/১১

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January

20

Tuesday

With this tech. also we can't accommodate enough users for comm.

In order to overcome in same tech, multiple access tech. were used instead of RF loop.

So TDMA, CDMA, FDMA.

- B/w issue reduced to some extent.

- There were limited no. of users we can reach so there was a limit.

$$\rightarrow 360 \text{ RF loop } 1000 \text{ users}$$

$$= 360000 \times 50 \text{ KHz}$$

$$= 18 \text{ MHz B/w}$$

also not possible

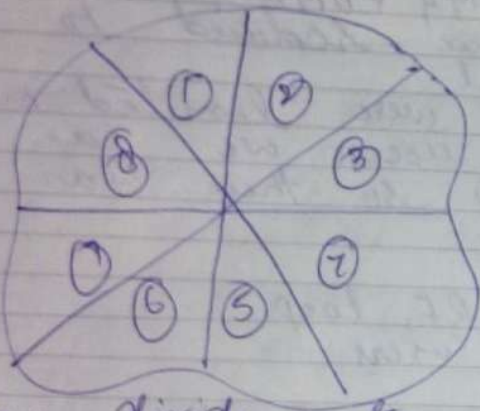
as B/w limited to 1 MHz or 2 MHz

21

January
Wednesday

(9)

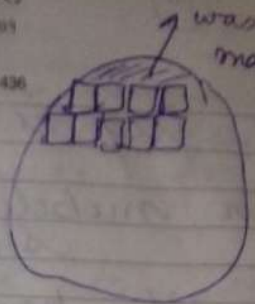
- Then the system shifted to cellular system (mobile).
- In cellular concept - study geo area to launch network, they divide the geo. area into smaller sub-areas



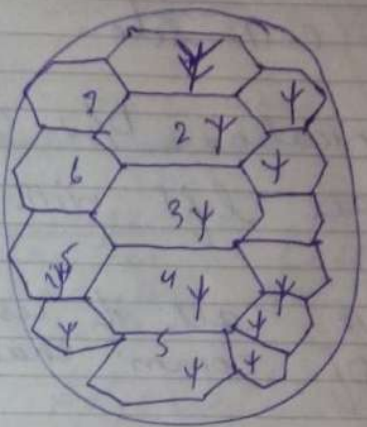
They divide the sub-area into cells.

Cell - small geo. unit of an area.

1. 2015-16
2. 2015-16
3. 2015-16
4. 2015-16
5. 2015-16
6. 2015-16
7. 2015-16
8. 2015-16
9. 2015-16
10. 2015-16
11. 2015-16
12. 2015-16
13. 2015-16
14. 2015-16
15. 2015-16
16. 2015-16
17. 2015-16
18. 2015-16
19. 2015-16
20. 2015-16
21. 2015-16
22. 2015-16
23. 2015-16
24. 2015-16
25. 2015-16
26. 2015-16
27. 2015-16
28. 2015-16
29. 2015-16
30. 2015-16
31. 2015-16



So, hexagonal shape is used for honey comb



Cell Structure

area ÷ into 7 structure

Cluster :- gp of cells (usually 7) where each cell using same RP tech or channel in single cluster.

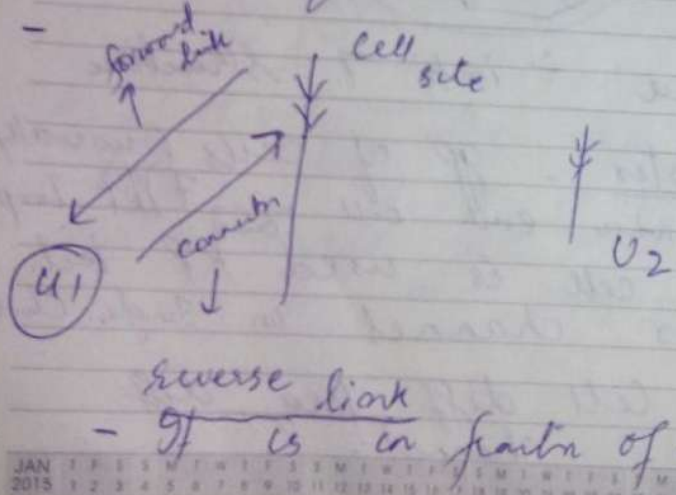
→ Cell diff freq, diff channel.

23

January
Friday

(10)

- No. of freq. is re-used.
- Cell - own mobile base station.
called mobile base station or cell site.
- Communication b/w cells:-
- No space left with cell site.
- When we call it uses pair of comm. channel.



Reverse link
- It is in pair of sec.

24

January
Saturday

24

- Cell site decides the transfer to pat cell.
- User to Cell site - forward link
- Cell site to user - reverse link
- No interference as each cell having diff frequency
- Re-use of freq used in another cell. depending they are not adjacent cells.
- Adjacent cell cant use the re-use freq. concept.
- We first drop first channel in order to go to the other channel.

Sunday 25

26

January
Monday

11

15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843

Pamela Day

→ When moving one cell to another during call, the cell MBS knows the user moves out of the coverage area. ∴ Sub is out of cov. area. It searches near cell site & sees strong freq & shift the call to that site/freq.

→ Shifting of call from one cell to another is called Handoff.

→ Due to this call is not dropped.

→ In handoff, call is transferred to other frequency.

26 NEW CHURCH 3843

15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843
 15 NEW CHURCH 3843

January
Tuesday

27

- B/w is efficiently used here but we hv to use the area efficiently.

→ It is very costly to construct the cellular structure of part area but it is a one time investment.

→ Out of reachable is bcz the user is in part of a cell not fully hexagonal as not connected to MBS no tower.

→ BSNL — strength of signal weak.

Splitting of cells.

In rural area cell of large size was used but in urban small size.

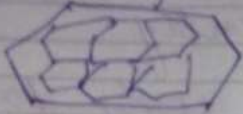
28

January
Wednesday

(12)

30 वास पुनराग 2015
 १ वास पुनराग पुनराग 2015
 मन्दी रा 2/4/5
 मन्दी - 7 राशि-उ-मन्दी 1430

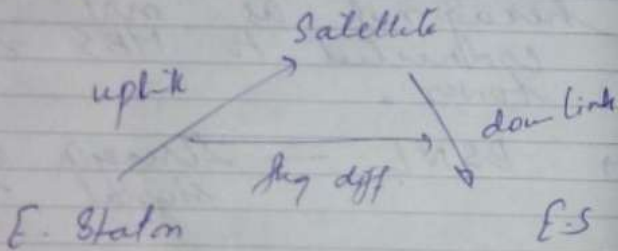
→ for freq. re-use there is splitting of cells.



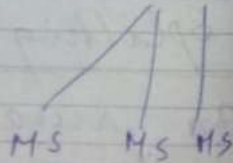
→ It ↑ no. of subscribers.

→ All ~~users~~ ^{Mobile Station} are connected to earth station.

The E.S. sends / connects to satellite, satellite send it back to earth station.



→ freq. of uplink is ↑ & ↓ for downlink.



Internet of

Things IOT

January

Thursday

29

30 वास पुनराग 2015
 १ वास पुनराग पुनराग 2015
 मन्दी रा 2/4/5
 मन्दी - 8 राशि-उ-मन्दी 1430

→ Command :-

ES sends code word to satellite, i.e. a command.

When E.S. receives command back, then it sends the signal to M.S.

This takes place in fraction of sec.

January
Friday

Requirement

repeat freq. after certain distance & we can accommodate more users.

→ Core channel interference
→ we hv to co-ordinate
as user is mobile.

- manage radio resource

1. Physical channel in GSM
2. Logical

2. Logical

125 - *freq*
6
124 - *1 January reserved*
Saturday 31 *Bard*

124 - freq

900 MHz band

1800 MHz band

BTS — Tower
Tower — BTS

BTS — Tower
Tower — BTS

MS — BTS

$$(0.90 \text{ H} - 0.15 \text{ H}_2)$$

BTS — Ms

(935 MHz - 900 MHz)

we have 25 MHz BW
in GSM.

04 (15) February Wednesday

Answer = 11
freq = Even
Min

২০ মাঘ বুধবার ১৪২১
৭ ফাল্গুন কৃষ্ণ শুক্লাব্দ ২০০৭
প্রতিপদ শের রাত ৯/০১
Hizri - 14 Rabi-us-sani 1436

There are 8 P. Channel /
carriers in GSM.

- 1 carrier $\rightarrow$ 8 channel

↓
carries
the data
voice & signa-
lling elements

$\rightarrow$ Time slot - via diff
info is send.

2) logical channel: diff
via info transmitting
this is C.C
In Physical Channel:

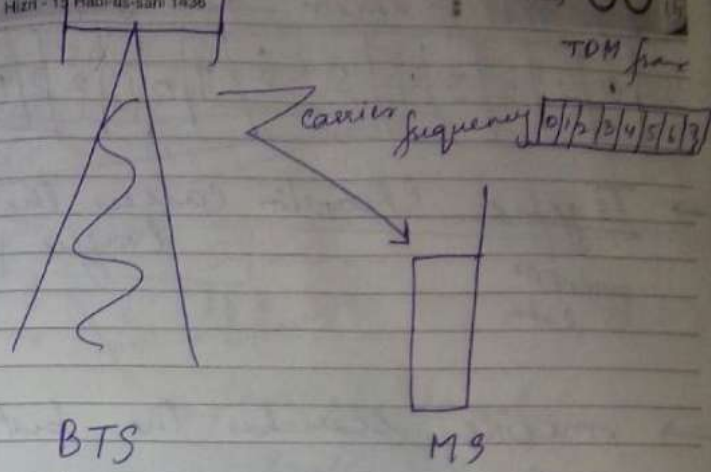
1. Traffic Channel
2. Logical Channel

FEB 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

২১ মাঘ বুধবার ১৪২১
২ ফাল্গুন কৃষ্ণ শুক্লাব্দ ২০০৭
বিহীরা আশ্বিন
Hizri - 15 Rabi-us-sani 1436

124 - carrier
freq

February 05 Thursday



for comm 2 diff devices
we need carrier freq

$\rightarrow$ frame (PCM) $\rightarrow$ 8 Time slots

$\rightarrow n - \text{frame} + 1$
 $= n + 1$

$\rightarrow$ TDM $\rightarrow$ divides the
carrier freq into
time slots

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 MAR 2015

16

Friday

TDM frame n

२२ भाषा शुद्ध्यात् १४२१

२ फाल्गुन कृष्ण शुक्रवार २०७१

দ্বিতীয়া ঘণ্টা ১২/০৮

16

२७-२८ भाषा शनि-श्रुति १८२१

३-४ फाल्गुन कृष्ण शनि-रवि २०४१

ତୃତୀୟା ସଂ: ୧୦/୧୬/ଚତୁର୍ଥୀ ସଂ: ୧୧

Hizri - 17-18 Rabi-us-sani 1436

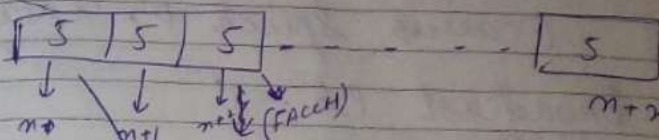
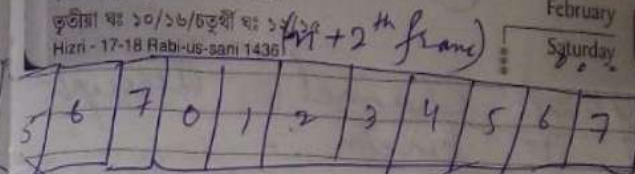
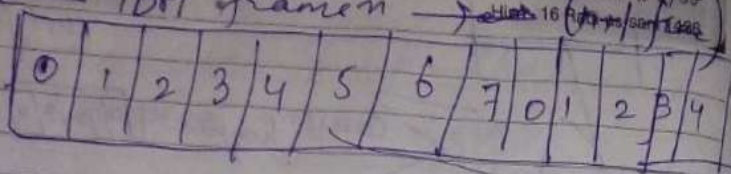
$$m + x^2$$

February

Saturday

07

1138 + 2th franc



Physical Channel /
Traffic Channel
(TCH)

→ I reserve the Channel for FCCH.

FCCH : logical channel
does carry data, voice.

→ GSM

4 types of Logical Channel

09 February
Monday

(17)

২৫ মাঘ সোমবার ১৪২১
৫ ফাল্গুন কৃষ্ণা সোমবার ২০৪৭
পঞ্চমী ষষ্ঠী ১/৫৫
Hizri - 19 Rabi-us-sani 1436

Logical Channel Disrupt & function :-

- 1.) Traffic Ch. (TCH)
(carries speech or data)
- 2.) Broadcast Ch.
eg BTS
(always in downlink only)
- 3.) Common Control Ch.
(CCCH)
- 4.) Dedicated Control (DCCH)

1) TCH :- carries data. If
get ÷ full rate or
half rate.

→ Permanent
or for some time slot
Traffic Channels are full

FEB 2015 S M T W T F S S M T W T F S S M T W T F S S M T
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

২৬ মাঘ মঙ্গলবার ১৪২১
৬ ফাল্গুন কৃষ্ণা মঙ্গলবার ২০৪৭
ষষ্ঠী ষষ্ঠী ০/১১
Hizri - 20 Rabi-us-sani 1436

February 10
Tuesday

Full rate or half rate

→ Full rate CH

TCH / FR & FD :-

data in GSM :- 13 Kbps/s
+ Channel coding
↓
extend to 22.8 Kbps/s

→ Half rate TCH / HR & HD :-

Speech sampled at half
the rate of full rate
speech

Channel. 6.5 Kbps + coding
results 11.4 Kbps
signal

2) Broadcast Ch :- always
transmitted on TS 0

(Time Slot)

S M T W T F S S M T W T F S S M T W T F S S M T
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 MAR 2015

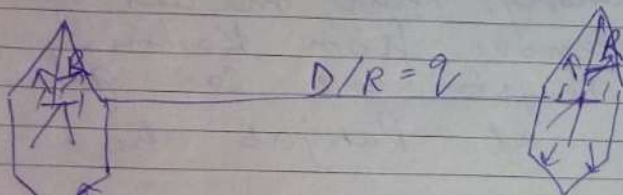
18 February (21) Wednesday

৫ মাঘুন বুধবার ১৪২১
১৪ ফাল্গুন কৃষ্ণা শুক্রবার ২০১৭
চাঁদমা ১৪ ৫/৫৪
Hizri - 28 Rabi-us-sani 1436

- MSCs are connected using optical fibre.
- Netwk Station subsystem
- Netwk management
- for monitoring BSS as well as MSS.

Date :- 18-8-2017

- To reduce interference.
- How we repeat frequency.
- Area ÷ into cells.



→ Cell - is radio-area (electro-magnetic reach upto what)

FEB 2016 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

৬ মাঘুন বুধবার ১৪২১
১৫/৭ ফাল্গুন কৃষ্ণা-শুক্র মঙ্গলবার ২০১৭
অমাবস্যা ১৫ ৬/১৭/৫৪৩৭৩৩ ০/৫৬
Hizri - 29 Rabi-us-sani 1436

February 19 Thursday

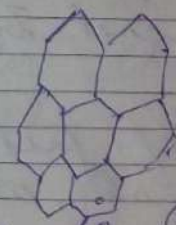
→ distance satisfy the repetition of frequency.

$$\rightarrow D/R = 7 \rightarrow \text{freq. reuse dist}$$

D: Distance of cell

R: Radius of a cell.

→ we hv to make cluster of cells.



7 cell cluster

→ Cells in cluster uses diff set of frequencies.

→ freq. allotted for a pair cell.

→ Diff freq - is a cluster

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

23 February Monday
 MSC - mobile switching centre (23)
 30 फाल्गुन सोमवार १४२३
 ५ फाल्गुन शुक्र सोमवार २०७१
 पक्षमी सा: १/२२
 Hizi - 3 Zamadilawal 1436

If system was in state $n-1$ & a new call arrives, system goes N .

→ Call departs = previous

→ Call block. if on talk is in on talk state & new call arrives then call block.

That probability is there.

→ Poisson distribution.

→ The chain is called Markov chain.

→ No. of users to calculate in system. We see the no. of cells in system.

Number of the jug is only in hundred in system.

- Grade of service.

24 February Tuesday
 ११ फाल्गुन मंगलवार १४२३
 ६ फाल्गुन शुक्र मंगलवार २०७१
 राशी स: ०/००
 Hizi - 4 Zamadilawal 1436

Cell A acc 1500 users

Cell B " 3000 users

- We hv 10 cells in system.

- MSC ~~can~~ how it can handle the users.

→ Cell - BSC - MSC

Total No. of users in system if we can face 10 cells are :-

$$G_t = \sum_{i=1}^{10} G_i$$

→ 50 × 8 = We can acc users.
 ↓ ↓ TDM
 No. of time slot 400
 jug.

25

February
Wednesday

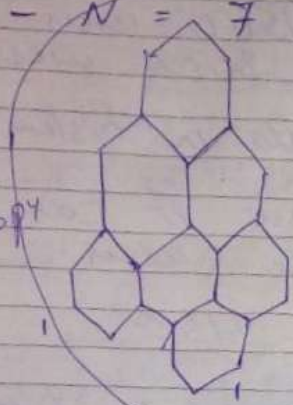
१२ काबून बुधवार १८२३
१३ फाल्गुन शुक्ल बुधवार २०७१
मङ्गशी अ: ८/२२
Hizri - 5 Zamadilawal 1436

Cluster

~~We hv to repeat d.~~

= Paste

1 Tier



- No need of next planning
bcz of clustering.
- Repitition of the cluster
so that full area is
covered.
- Cell - around 6 co-ches
in 1 Tier if shape is
hexagonal
- Tier 2 - it is in large
dist than Tier 1.

the 6 cells

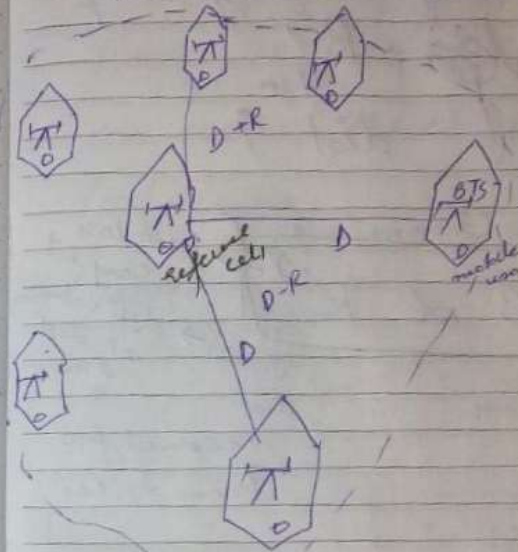
over the

the parent cell is (cell 1)

१७ शाबुन वृहस्पदिवात १४२१
८ फाल्गुन शुक्ल गुरुवार २०७५
अश्लेषा रा: ७/७४
Hizri - 6 Zamadilawal 1436

February 26
Thursday

→ Tier 2 at large dist so no more interference



Other
cells
uses
diff
freq

- Int only
if same
freq

- We see when the mobile is at the edge of cell it get interf. b/w of neigh cell.
- Distance b/w the cells r diff.

27 February Friday

(26)

18 फाल्गुन शुक्रवार १४३३
९ फाल्गुन शुक्रवार २०७१
मकर संक्रांति ०८/०२
Hizri - 7 Zamadilawal 1436

- free space comm, the power that vary as a funct of $R^{-1/2}$ when
Power $\downarrow$ dist $\uparrow$

→ Mobile Comm. $\left[\begin{array}{c} C \\ \downarrow \\ \text{power} \end{array} \right] \left[\begin{array}{c} \propto \\ \downarrow \\ \text{dist} \end{array} \right] \left[\begin{array}{c} R \\ \downarrow \\ \text{mean dist for tra to rec} \end{array} \right]$
path loss component
→ diff value at diff area.

$\gamma \rightarrow$ environ dependent

$$C = \alpha R^{-\gamma}$$

α is constant of proportionality.

February 28 Saturday

18-19 फाल्गुन शनि-रवि १४३३
१०-११ फाल्गुन शुक्र-शनि २०७१
मकर संक्रांति ०८/०२/१४३३ ०८/०२
Hizri - 8-9 Zamadilawal 1436

→ If D is not managed prop, signal is received from diff cells (6) for a user, there are actually

Interference.

→ 1 user can hv 6 Int. signal

→ Carrier power = Performance

Interference

$$\frac{C}{I} = \frac{R^{-\gamma}}{\sum_{i=1}^6 D_i K^{-\gamma}}$$

$$\frac{R^{-\gamma}}{6} = \text{Perf}$$

Sunday 01

FEB 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 MAR 2015

March
Monday

১৭ ফাল্গুন সোমবার ১৪২২
৭২ ফাল্গুন শুক্ল সোমবার ২০৩৭
বাদশী নং ৫/২২
Hizri - 10 Zamadilawal 1436

$$\rightarrow \frac{C}{I} = R^{-\gamma} \sum_{k=1}^L D_k^{-\gamma}$$

÷ by R

$$\frac{C}{I} = \frac{1}{\sum_{k=1}^b (q_k)^{-\gamma}}$$

$$I_k = \frac{D_k}{R}$$

where q_k is the co-channel
interf. reduction factor

k - represents k^{th} cell &

$V \rightarrow$ propagation path

MAR	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T
2015	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31

१८ काष्ठानु मन्त्रालय १८२१
१३ फाल्गुन शुक्ल मंगलवार २०७९
जयप्रान्तीय शांति ७/७२
Hizri -11 Zamadilawal 1436

March
Tuesday 03

i.e. calculated by the
turban (diff area diff)

$$\frac{C}{I} = \frac{R - r}{G D - r}$$

by R

$$\frac{C}{I} = \frac{q^r}{6}$$

$$q^v = 6 \frac{C}{I}$$

for N size Cluster

$$q = \sqrt{3} N$$

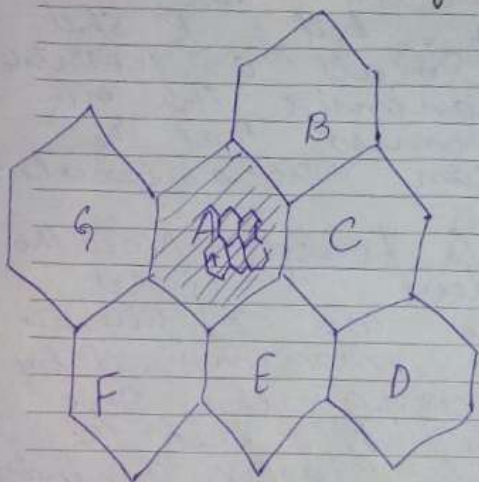
$N = 7$ Size Cluster.

W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	* APP
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	*

২৮ ফাযুন শুকান ১৪২২
৩/৮ বৈত্র কৃষ্ণ শুকান ২৪৩৩
সপ্তমী ত্রা: ৬/২২/৪৪মী শে: ত্রা: ৩/৪৩
Hizri-21 Zamadilawar 1423

determining the value of N , another trade off is encountered in that N must be small to acc. large no. of subscribers but shd be sufficiently large to minimize the interference

1. Cell splitting:-


$$N=7$$

radio waves

March 14
Saturday

२२-७० ग्राह्यन सनि-रवि १८२१
२-१० वैश्व कृष्ण सनि-रवि २०७५
नवमी एकादशी ८/२३/५१ नवमी रा: २/७२
Hizri - 22-23 Zamadilawal 1436

Cell A split into smaller cells
mostly done on edges
- clusters are in 1 cell

- mostly done on edges
- Clusters are in 1 cell

Cell — micro-cell

- macro-cell
- larger cell

- Small towers in these small micro cells.

- radius is small.
- No. of freq. $\uparrow$, Capacity $\uparrow$ but freq. re-use factor $= 1$.

but conventionally it
was = 7.

Sunday 15

- we design a low power antenna to make it possible. The signal is limited to a part cell. So, interference can be controlled.

✓ also radio-area is

16 March
Monday

(32)

১ চৈত্র সোমবার ১৪২১
৭৭ চৈত্র কৃষ্ণ সোমবার ২০৭৭
একাদশী রা: ১৩/৪৮
Hizri -24 Zamad/lawal 1421

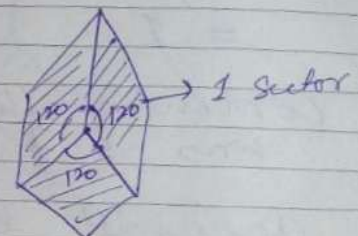
- micro-cells can not be worked in congesting area. We $\rightarrow$ micro cell into nano ~~area~~ cell.

2.) Cell Swallowing:

One cell $\div$ into sectors

total area covered by cellular area = 360

360 is $\div$ by 6 or
whatever depends on our



antenna placed in these sectors.

9m. normal cell, antenna is placed as omni-direction — radiates the electro mag. signal in every direction

२ छत्र मण्डलवार १८२१
१२ वीर कृष्ण मंगलवार २०७१
दान्दी रा: १०/८३
Huzri -25 Zamadilawal 1436

March 17
Tuesday

- but creates problem at adjacent cell.

- Instead of using omni-directional we use directional antenna i.e. Only for a particular direction. So, no signal transmitted And interference ↓

- Each sector has own set of freq.

No. ↑, Capacity, ↓
int. ↓

Date : 24-8-17

Cell Strength & Parameters:

Hand off In a Cellular System

- The strength of mobile comm depends on how much power or signal strength is received by MS from the cell (parent) as well as

18

March
Wednesday

(33)

৩ চৈত্র বৃহস্পতিবার ১৪২১

৭৩ বৈশাখ শুক্লা ২০৪৭

ব্রহ্মোদিশী রাঃ ৮/৩১

Hizri - 26 Zamadilawal 1436

Beruni

from the adjacent cells.

- depend contour of the cell
+ power handling capacity

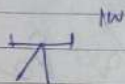
i) How much power is received from by the base station

ii) Contour of cell.

iii) location of MS.

- centre of cell
- edge of cell

→ received max from own base station.

power varies as
of func of distance

-60dB

-70dB

→ power loss - more dist.
→ y in dB bcz we always

MAR 2018

৪ চৈত্র বৃহস্পতিবার ১৪২১

৭৪ বৈশাখ শুক্লা ২০৪৭

চতুর্দশী সঃ ৬/০৭

Hizri - 27 Zamadilawal 1436

March
Thursday

19

- linear scale
- dB "

measure power w.r.t reference

→ actual power = $\frac{\text{received power}}{\text{max power (Pm)}}$

→ when we take the log of this, this changes to dB.

$$\rightarrow \frac{1}{1} = \log 1 = 0 \text{ dB}$$

↓
max power

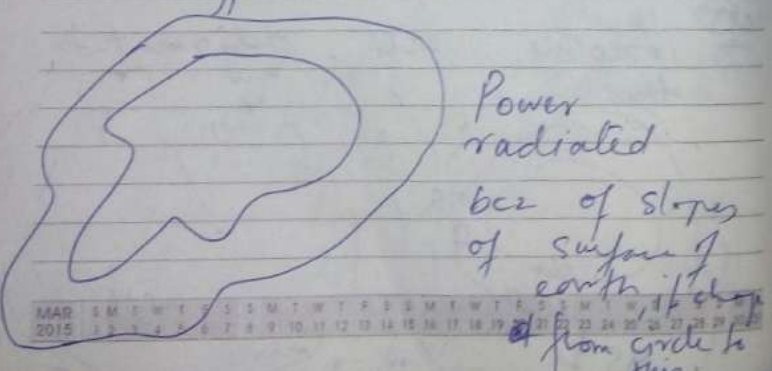
→ So -ve bcz we always get the less power as compared to received power.

$$-60 \text{ dB} < -70 \text{ dB}$$

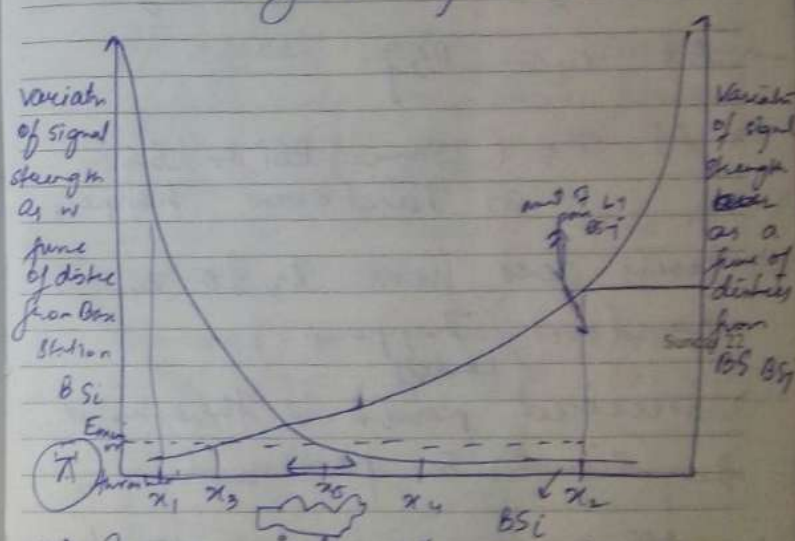
→ Another cell adjacent to this



- The MS will receive power from BS_j or BS_i .
- When BS_i moves it enters to BS_j . (mobile station) Max power receive when it latches.
- from 2 BS's power is equal what will happen, BS_i or BS_j will serve?
- Power radiated as co-centric circles.
- diff power strength at diff statn.



- This phenomena i.e. moving MS to another cell i.e. goes from parent cell to adjacent cell is called Hand off.
- Power varies exponentially as a function of distances.



- Suppose MS at x_1 , location x_1 it receives Max power from BS_i . BS_i serves it.

March
Monday

receives if power is above than threshold

Hizn-2 Zamad us-sani 1438

it gives more power as compared to BS.

- If at x_2 , $BS_i < \text{so served by } BS_j$.

→ Mobile user is static & reach at x_2

→ At x_3 , it is served by BS₁.

→ Af xy — BSj

→ At x_5 (same) BSi & Bg
so hand over happens

→ users goes from x_3 to x_4 ,
handovers / happens.

→ received ^{power} ~~threshold~~ > threshold.
then, signal is processed.

→ MS always monitor this.

→ freq. allotted by BS; starting

MAR	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	
2018	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

५. ऊ.प्र. अभिलेखांक १४२१

५ जेष्ठ शुक्ल मंगलवार २०७२

চতুর্থী খঃ ৭/৪০

Hizni - 3 Zamadius-sani 1436

March

24

but when goes to BSj we monitor the power not frequency.

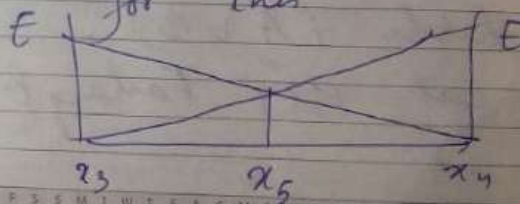
→ Ping Pong effect: MS
movement is random
freq. change continuously
It shd be avoided.

→ PP effect at x_5 . (moving from x_5 & way back.

Avoid :- PP effect.

- Battery is drained due to PP
- Handoff :- new freq is allotted

- network operator decide what from served BS; a BS_j & the parameter we use for this



March
Friday

(37)

Annapurna Puja

১২ চৈত্র শুক্লাবাত ১৪২১
৮ চৈত্র শুক্লা শুক্লাবাত ২০/১২
অষ্টমী শো: রা: ৫/২৩
Hizn-6 Zamadius-sani 1438

360

$$a = 1.7$$

$$A = \frac{360}{360}$$

$$a = \frac{30}{3600} \text{ sec}$$

$a = 3$ carlang.

MAR 2015	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

૧૭-૧૮ ઈશ્વર જ્ઞાન-રવિ ૧૪૨૧
 ૧-૧૧ દેવ શુભ જ્ઞાન-રવિ ૨૦૭૨
 નવમી અશ્વિન/નવમી વ્રા: ૭/૧૦
 Hizri - 7-8 Zamadi-usani 1436

Date: - 8-09-2017

March

Saturday

28

Rannabari

Hand-off : ~~cont~~

Types of handoff

4- Types:

- BSC, MSC & within the cell (change)

we hv handoff

- When boundary crosses hand off.

1) Intra Cell - Intra BSC
handoff :-

Subscriber in the cell
& within own BSC (base
station controller)

- BTS changes the freq. of subscribed when a user do not hv the way to hv a freq.

- old freq assigned to other user

- user migrated new
old to new jug the
cs hands off

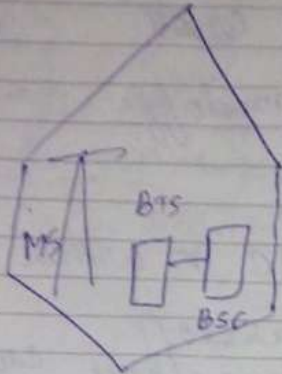
[illegible]

30

March
Monday

cell (38)

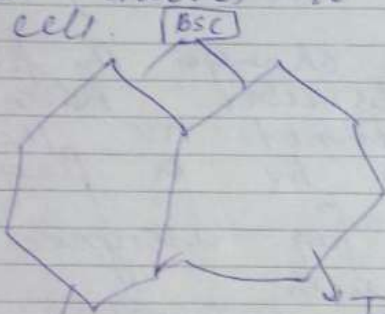
38 दिवस शनिवार 18/03/2015
 40 दिवस बुधवार 20/03/2015
 एनपी 42 4/25
 Hazi - 9 Zamadus-sani 1430



Old Channel
New Channel

2) Inter Cell - Intra BSC Hand off:

When moves to another cell.



Transfer to new cell.

actual serving cell

38 दिवस शनिवार 18/03/2015

40 दिवस बुधवार 20/03/2015

एनपी 42 4/25

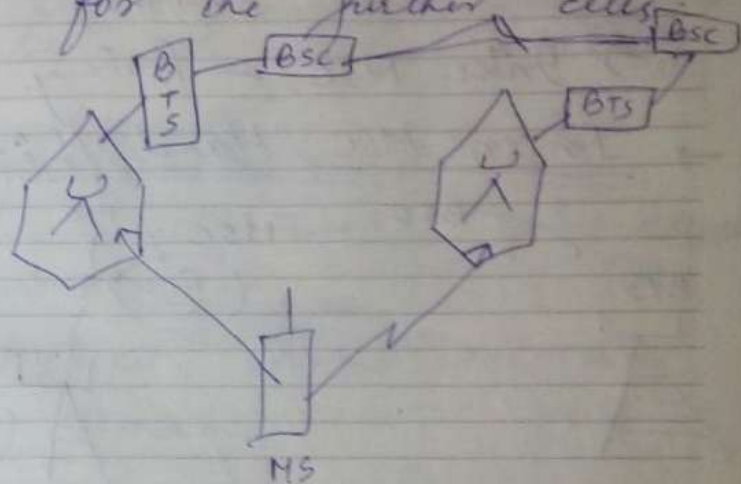
Hazi - 10 Zamadus-sani 1430

March
Tuesday

31

3) Inter Cell - Inter BSC H.O.:-

Where two cells & served by diff BSC. As at a certain distance we don't have same BSC for the further cells.



- Actual decision for a hand over to take place is the (MSC) mobile switching center.
- BTS only monitor the transfer.

Switching + DB + Billing
→ MSC

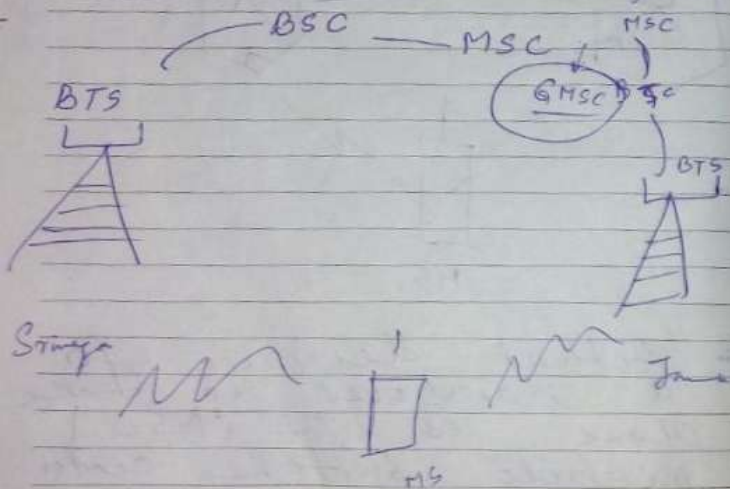
39
01

April
Wednesday

Yearly Bank Closing Day

১৭ চৈত্র সুববার ১৪২১
৭২ বৈশাখ শুকল ২০৫২
শ্রাবণী ৩১ ১০/৪৬
Hizri-11 Zamadi-us-sani 1436

- Handover by the MSC is not possible, so roaming is not possible
- Gateway MSC (in Chandpur) (from 1 zone to another)
- Inter MSC - Gateway
- Inter MSC Hand off :-



Date:- 9-9-2017

১৮ চৈত্র সুববার ১৪২১
৭৩ বৈশাখ শুকল ২০৫২
শ্রাবণী ৩১ ১২/৪৬
Hizri-12 Zamadi-us-sani 1436

April
Thursday 02

Mahebir Jayanti

Channel allocation:-

allot limited freq for all traffic

- Base Station manager
- Starts at 633 initially freq after FDM but later on we did clustering

$$\frac{413}{7} \leftarrow \text{Total freq} = 59 \text{ freq}$$

for 1 cell

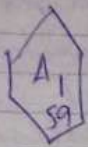
cluster Techniques.

- 1) fixed channel allocation
- 2) Dynamic "
- 3) Hybrid channel all.

A set of channel is permanently allotted to each cell in system.

03 April Friday
 15 ऐब सन्वत् 2003
 १४ वैशख शुक्ल शनि-रवि ०२
 छठमी १३ ०/०३
 Hizi-13 Zarnadus-sani1436

→ 9m Cell 1 → 1 set of
 freq.



If all freq. are
 not available
 & new call
 arrives.

He uses the unused
 freq. i.e. Simple
borrowing scheme from
 adjacent cells.

Cell is called —
 donor cell
 from borrowed

— preference seen by
 seen by dist
 to avoid, adjacent
 cell interference

20-21 ऐब सन्वत् 2003
 १५ वैशख शुक्ल शनि-रवि ०२
 छठमी १३ ०/०३
 Hizi-14-15 Zarnadus-sani1436

04 April Saturday

— ~~Other~~ The borrowing
 freq. must be
 given back to donor
 cell.

→ Static technique

↓
 complex borrowing scheme

↓
 channel ÷ into groups.

1 grp — reserved
 kept

1 grp — ÷ into cells

2) Dynamic CA :-

we create center
 pool.

— No resource wasted
 as freq. not permanently
 allot

06 April Monday

- center of pool - all
- If needed they request to the pool.
- Algo is complex.

3) Hybrid: static + dynamic (adv)
↓
Simplicity

- channel into gps
- 1 gp — priority allot
- 2 gp — center po
- all cells are exhausted
then they request center pool. We hv a scheme

APR 2015

07 April Tuesday

- to allocate freq when we are moving highway. Then from a cell we borrow freq.
- We borrow from D+1 channel/att such that no channel adj. interference.

Date: 16

GSM architect

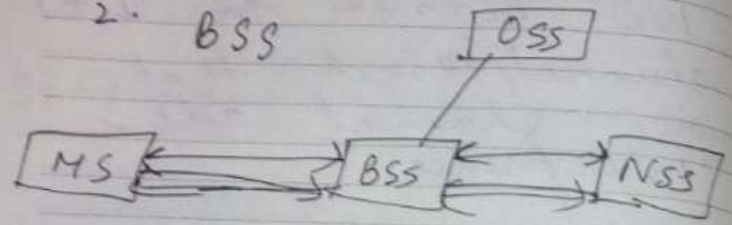
- first arch — given by CPAT (European org.)
- CPAT developed TDMA
- It is stand alone system via out Europ.
- Start 1982
- 1992 → germany commercialize the

MAY 2015

GSM architecture called D

- Basic components are

1. MS
2. BSS
3. NSS



MS: mobile station / standalone piece of equipment for certain services

- Can be conn to external device

- Parts

1. Mobile equipment
2. Sim

part of MS $\subseteq$ consists of i) IMSI ii) TMSI
(iii) IMEI

IMSI :- Call is always encrypted form i.e IMSI

To further enhance the security code is not assigned temporary

i.e TMSI (Temporary mobile No)

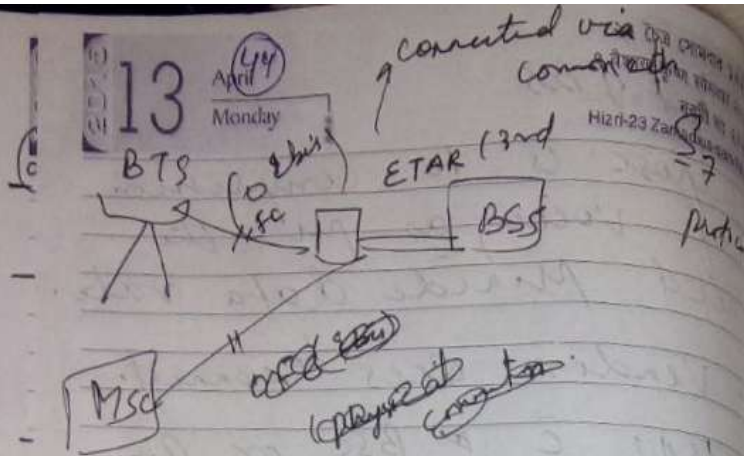
IMEI - No. $\subseteq$ equipment identity

→ These all are independent on SIM

- MS is network independent

- $\subseteq$ makes network dependent is SIM

- Mobile is an equip without SIM



- S₇ protocol adopted by every manufacturing companies.

(3) - NSS :- network switch subsystem
GSM - intelligent network system

Main funt of NSS is connect GSM users with other users (ISDN facilities)

14 April Tuesday

MSC → call set-up.

mobile identifi / managed is done by MSC.

(4) IWF :- Gateway MSC
Inter networking function.

→ J&K MSC's connect to Delhi MSC's

(a) HLR :- Home location register.

- Standalone computer. core switching capabilities.

- Store db.

- Subscriber info is there & also store current location but not the actual location of subscriber.

15 April Wednesday
Bengali New Year's Day
১৫ বৈশাখ কৃষ্ণা শুক্লাব্দ ১৪৩৬
একাদশী ৫:১১/৫:০৮
Hizri-25 Zamadus-sani 1436

→ AUC authentication Centre
↓ manages the security data for subscribers authentication.

(v) EIR: equipment identity register

store IMIE of mobile equipment

data MSC → VLR → visitor location register

— update addresses / info of user.

VLR — temporary stores subscriber data currently served by MSC

16 April Thursday
১৬ বৈশাখ কৃষ্ণা শুক্লাব্দ ১৪৩৬
দ্বাদশী ৫:০৮/৫:০৯
Hizri-26 Zamadus-sani 1436

HLR update = VLR

vi) → Gateway MSC: In order to setup a request call, call is initially routed to a gateway MSC. It finds the correct HLR of user in C MSC. Then it connects to it.

(vii) → Signalling transfer point: STP concerned in the interface. It optimize the cost. (include balance speed & cost).

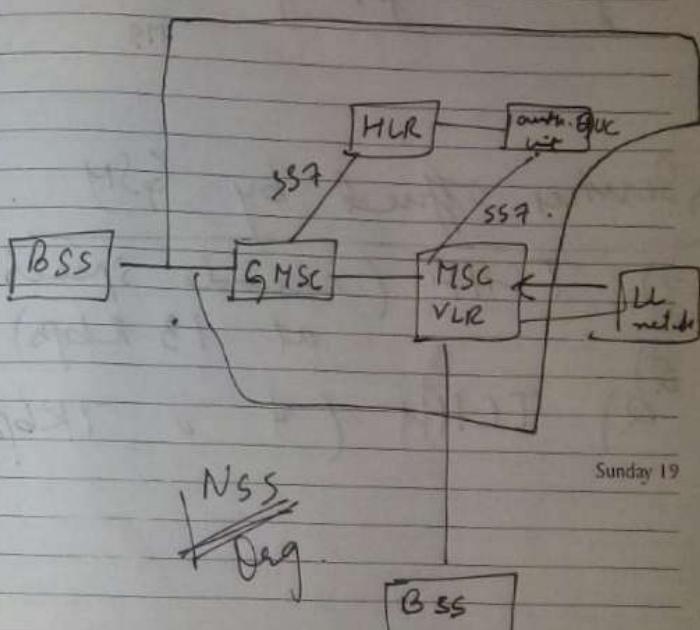
MSC/VLR/GS /HLR all interfaces optimize

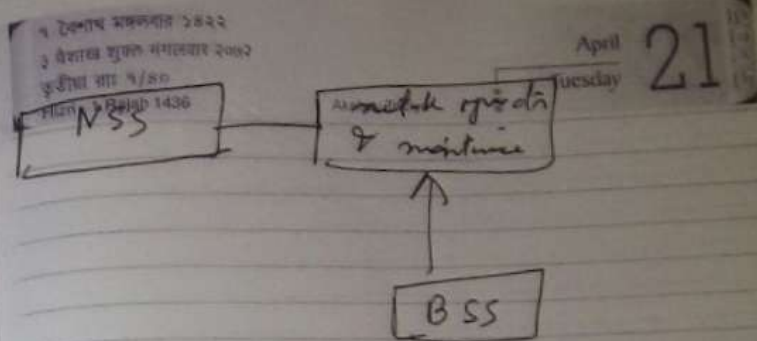
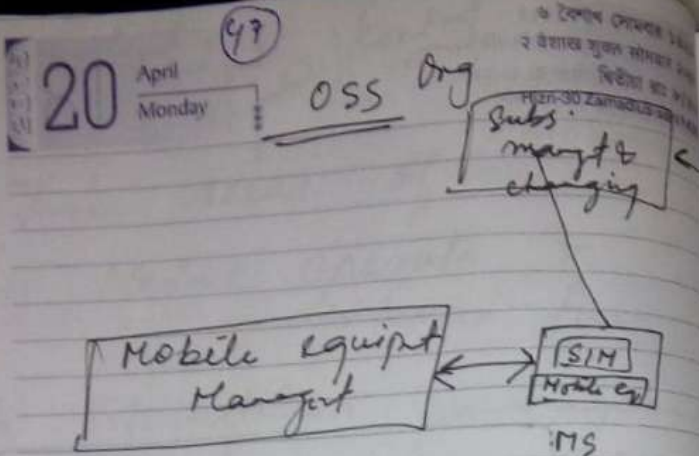
APR 2015 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 MAY 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

(Front end part of mobile) (Control room) (46)
 17 April Friday
 operated
 Sub
 Hizri-27 Zam
 → These areas of OSS

- 1) Network operation & maintenance function
- 2) Subscription management including charging & billing
- 3) Mobile equipment management. (configuration)

Functional Architecture & Principle Interfaces





Q → Services offered by GSM

- 1) TCHF (encode speech at 13 Kbps)
- 2) TCHH (" " 7 Kbps)